TIME-OF-FLIGHT DISTANCE SENSOR

ARK DIST

DroneCAN and MAVLink laser distance sensor in short-, mid- and long-range variants

EXPOSED INTERFACES CAN UART SWD

The USA-built ARK DIST pairs a Broadcom AFBR-S50 time-of-flight sensor with an STM32F412VG. The SR, MR and LR variants share one board and case and differ only in the sensor module.



01 KEY FEATURES

- Broadcom AFBR-S50 time-of-flight sensor, 850 nm Class 1 laser
- Max reported distance: SR 30 m, MR 50 m, LR 100 m
- DroneCAN and MAVLink 2 output
- 2× CAN and UART on Pixhawk-standard JST-GH, debug on JST-SH
- PX4 and ArduPilot; ROS 2 via ros2_dronecan

02 SPECIFICATIONS

SENSOR	
Time-of-flight module	Broadcom AFBR-S50, see 03
Laser	850 nm, IEC 60825-1 Class 1 (eye-safe)
Transmitter beam	2° × 2°, 3.5 cm spot at 1 m
Receiver (SR, MR)	32 pixels, 12.4° × 6.2° field of view

INTERFACES	
CAN	2× 4-pin JST-GH, DroneCAN
UART	6-pin JST-GH with RTS/CTS, MAVLink 2
Debug	6-pin JST-SH, SWD and console

PROCESSOR	
MCU	STM32F412VG

ELECTRICAL & MECHANICAL	
Supply voltage	5 V
Supply current	78–86 mA
Dimensions	20 × 28 × 14 mm
Weight	4 g

AT A GLANCE	
Sensor	Broadcom AFBR-S50
Variants	SR / MR / LR
Max reported distance	30 / 50 / 100 m
Update rate	20 / 15 / 10 Hz
Supply	5 V
Dimensions	20 × 28 × 14 mm
Weight	4 g
Hardware rev	Rev 1.0

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- IEC 60825-1 Class 1 laser product

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-dist-sr/
arkelectron.com/product/ark-dist-mr/
arkelectron.com/product/ark-dist-lr/

DOCS docs.arkelectron.com/products/sensor/ark-dist

CAD github.com/ARK-Electronics/ark-hardware

03 VARIANTS

	ARK DIST SR	ARK DIST MR	ARK DIST LR
Time-of-flight module	AFBR-S50LV85D	AFBR-S50LX85D	AFBR-S50EX85D
Update rate	20 Hz	15 Hz	10 Hz
Max reported distance	30 m	50 m	100 m
Min reported distance	0 m	0.1 m	0.6 m

04 RANGE

SURFACE	ARK DIST SR		ARK DIST MR		ARK DIST LR	
	BRIGHT SUN	DUSK	BRIGHT SUN	DUSK	BRIGHT SUN	DUSK
Asphalt	18 m	30 m	18 m	40 m	18 m	40 m
Grass	18 m	30 m*	18 m	50 m*	40 m	82 m

Reliable range: the farthest distance with ≥50 % of readings valid and within 5 m of the reference, at default settings. Bright sun: sun 49–61° above the horizon; a lower sun gives more range. Dusk: sun below 6°. * Limited by the maximum reported distance; the sensor was still measuring at the flight ceiling (SR 38 m, MR 54 m). Wet surfaces reduce range; open water (~2 % reflectivity) gives only short, intermittent returns.

05 CAN CONNECTOR (×2)

PIN	SIGNAL	VOLTAGE
1	5V	5.0 V
2	CAN_P	5.0 V
3	CAN_N	5.0 V
4	GND	GND

Connector: 4-pin JST-GH, Pixhawk standard. Chain additional nodes from the second CAN connector.

06 UART CONNECTOR

PIN	SIGNAL	VOLTAGE
1	5V_UART	5.0 V
2	USART3_RX	3.3 V
3	USART3_TX	3.3 V
4	USART3_RTS	3.3 V
5	USART3_CTS	3.3 V
6	GND	GND

Connector: 6-pin JST-GH, Pixhawk standard.

07 DEBUG CONNECTOR

PIN	SIGNAL	VOLTAGE
1	3.3V	3.3 V
2	USART2_TX	3.3 V
3	USART2_RX	3.3 V
4	FMU_SWDIO	3.3 V
5	FMU_SWCLK	3.3 V
6	GND	GND

Connector: 6-pin JST-SH, Pixhawk standard.

ARK-DS-ARK-DIST | Issue 2026-10 | Compiled from ARK Electronics published documentation and ARK flight measurements. Laser and optical characteristics per Broadcom.
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